

to produce sound, requiring no external power supply.

Nowadays, there are various techniques to harvest this energy from the surrounding environment. These techniques can be easily applied for compact low-power devices where methods like solar power generation are not viable. The power harnessed by these techniques is enough to power small, wireless autonomous devices, like those used in wearable electronics and wireless sensor networks.

Let us have a look at some popular energy harvesting methods and understand how exactly they harvest energy.

Vibrational energy harvesters

If a low-power design demands on-chip energy harvesting solution, then vibrational energy harvesters are perfect for use, especially for wireless nodes. Vibrational energy harvesters derive power from physical vibrations by using micro electromechanical systems (MEMS). It is the most suitable technology to realise IoT-sensing nodes because on a single chip (or a package), this technology enables integrated fabrication of sensors/actuators, electronic circuits for processing information and radio frequency communication, antennas, and energy harvesters.

For using vibrational energy harvesters as a power source, the average power consumption must be tuned around $100\mu\text{W}$ for a reliable operation. For wireless systems, this tuning can be done by reducing the transmitting duty ratio.

As mentioned, these harvesters take advantage of mechanical energy present in the surroundings. The surrounding mechanical vibrations in the environment have a particular dominant frequency. When the fabricated MEMS oscillator is tuned to this dominant frequency, we can expect to see power generated.

This concept is similar to the impedance-matching concept in which for maximum power transfer, we match the load with the output impedance of the circuit. Such a mechanism is usually referred to as the Velocity-Damped Resonance Generator (VDRG), as the velocity of a physically exciting mass is mechano-electrically coupled to deliver electrical power to the external load.

Fig. 1 shows the power conversion mechanism in vibrational energy

Problem with Batteries

With compact lithium or nickel cadmium batteries being currently used in almost all portable electronic devices, certain problems faced while using them do not quite make batteries the best choice. Designers are seeking solutions to either eliminate the need for energy storage in devices or use an alternative energy. Listed below are some problems associated with batteries and how the challenges faced can be overcome.

1. Weight and space. Batteries can be held accountable for most of the weight of a portable electronic device. In some cases, important circuitry is sacrificed to accommodate batteries. Features and specifications are limited if the device space is constrained for accommodation of batteries. PCB layout also becomes complicated, and wiring has to be done in a way that power rails do not introduce noise in the circuit. Using a harvester circuit on chip, designers efficiently manage the layout and avoid noise related issues.

2. Cost. The cost of batteries is quite significant in comparison to other components in a device. Energy harvesters are easily integrated in wireless systems and many other applications without affecting the cost much.

3. Safety. Lithium-ion and lithium-metal batteries are prone to thermal runaway, a process in which internal shorts lead to significantly high temperatures that can lead to fire and explosion. If there is a fault in manufacturing, or if proper care during charge and discharge is not taken for the batteries, it may lead to a fire. Though this is more acute in large-scale applications, it may occur in small-scale products when batteries are charged/discharged without proper management and protection circuits.

4. Deep discharge. Deep discharge, or over discharge, occurs when the battery is discharged at a very high rate. Battery manufacturers specify the cut-off voltage below which, if discharged, the battery can get permanently damaged. It is quite difficult to recharge at full capacity thereafter, because the internal resistance of the cell rises significantly. The cut-off voltage is extremely sensitive to temperature, so care has to be taken when the battery is operated under such conditions.

5. Cycle life. The life cycle refers to the number of charge and discharge cycles that a battery can complete before it degrades. Lithium batteries can last for a few hundred charge and discharge cycles, after which they lose their performance and need to be discarded.

6. Management systems. Batteries or battery packs may require management systems for validating whether the cells are working in safe operating conditions. Though the battery management circuits are expensive and take up lot of space in a device, they are incapable of fully preventing battery failure reactions that happen rapidly.

7. Power density. Batteries take a huge time to charge and have various issues regarding discharge rates. This brings the need for alternatives like supercapacitors, which can charge/discharge substantially faster than batteries.

8. Maintenance. Batteries need to be replaced after they have degraded. Such replacement in remote locations is difficult. With the use of battery-less solutions, or supercapacitors, such issues are eliminated.

9. Pollution. Since lithium batteries need to be disposed of after a few hundred charge and discharge cycles, recycling these batteries, or any other batteries, is difficult. This adds to toxic waste in the environment. Batteries use toxic materials such as lead, cobalt, and fluorinated electrolytes in manufacturing. Most of the electrolytes used in lithium-ion batteries are super halogens, and the vast majority of these electrolytes contain toxic halogens. Moreover, gas emission from malfunctioning lithium-ion batteries creates highly toxic emission.

harvesters. The ' Q_{in} ' and ' Q_{ex} ' refer to the internal and external quality factors, respectively. Large internal losses are present when Q_{in} is small, since the mechanical resonance is not efficiently excited and there would be a low output if there is insufficient coupling between mechanical and electrical systems when Q_{ex} is large. Similar to impedance matching, the maximum power at the output can be obtained if $Q_{in} = Q_{ex}$.

Below are a few examples of energy harvesting devices that include the concept of mechanical energy harvesting.

Electrostatic devices. In the

presence of an electric field, when mechanical motion of a conductor occurs, electrostatic current is induced. The electrostatic devices use permanent electric charges or electrets that build the electrostatic fields by trapping electrons or ions in electrical insulator materials. Electrets generate internal and external electric fields and can store charges for a prolonged period of time (say 100 years).

Electromagnetic devices. This electromagnetic harvester follows Faraday's law that induces an electric field by the changes in magnetic field. The mechanical motion of permanent magnets and coils induces electric power,